Dissolved Carbon Dioxide

Prodigy Anesthesia

Like oxygen, the amount of carbon dioxide dissolved in the bloodstream can be calculated. An important fact to remember is that carbon dioxide is more than twenty times more soluble in blood than oxygen. To calculate the amount of CO2 dissolved in the blood, multiply the PaCO2 by 0.067.

Calculate the amount of CO2 dissolved in the blood if the PaCO2 is 44 mmHg.

Dissolved CO2 = PaCO2 X 0.067

Dissolved CO2 = 44 X 0.067

Dissolved CO2 = 2.95 mL/dL

Dissolved Carbon Dioxide Calculation Practice Problems

Calculate the amount of CO2 dissolved in the blood if the PaCO2 is:

1. 40 mmHg
2. 55 mmHg
3. 29 mmHg
4. 62 mmHg
5. 46 mmHg

Dissolved Carbon Dioxide Content Answers

(answers are in mL/dL)

1. 2.68
2. 3.69
3. 1.94
4. 4.15
5. 3.08